

Arjan Hayre

Software Engineer | C++ & Python | Backend & Systems | Applied AI
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Software engineer and MEng graduate in Electronic and Information Engineering from Imperial College London, with three software engineering internships and experience across backend services, full-stack applications, real-time systems and applied AI. Strong in C++ and Python, with practical experience in software design, testing, performance optimisation and end-to-end delivery.

Education

Imperial College London

Master of Engineering, Electronic and Information Engineering

MEng project: Design and Evaluation of a Local LLM Game-Master System for a Non-Linear Unreal Engine RPG.

Oct 2022 – Jun 2026

London, UK

Experience

Advanced Robotics and Computing Ltd

Founder

Feb 2025 – Present

London, UK

- Founded a software and applied-AI venture; developing Invoice Capture, a B2B e-invoicing and document-automation platform for construction workflows.
- Own product architecture and end-to-end implementation across Python, TypeScript, React, PostgreSQL, Supabase, APIs and AI-assisted document processing.

Xcceler8 Ltd (Client WeBook)

Full Stack Engineer

Jun 2024 – Sep 2024

Riyadh, Saudi Arabia / Remote

- Reduced median API response time from approximately 180 ms to 95 ms under peak load by introducing Redis caching and redesigning PostgreSQL schemas.
- Supported booking and reservation workflows in Spring Boot microservices handling hundreds of thousands of HTTP requests per day by implementing Camunda BPMN workflows.
- Improved API gateway resilience by adding rate limiting and request validation to control invalid traffic.

Xcceler8 Ltd

Full Stack Engineer

Jun 2023 – Sep 2023

Remote

- Improved login stability for thousands of daily users by integrating Firebase Authentication and Firestore with backend services and implementing OAuth2/JWT flows.
- Enforced role-specific permissions across the application by developing role-based access-control modules.
- Enabled support teams to manage users and operational workflows by building React and Material UI admin-dashboard components.

Dorchester Homes

Developer Intern

Jul 2022 – Oct 2022

London, UK

- Delivered a Python property-valuation model from stakeholder requirements and data preparation through testing and interpretation.
- Built a project-tracking and VAT-automation dashboard used weekly by operations staff, automating manual monthly reporting steps.
- Optimised SQLite schemas and queries, reducing average report-generation time from approximately 2.1s to 0.7s; coordinated rollout with operations stakeholders.

Projects

Local LLM Game-Master System – Unreal Engine 5

MEng Project, 2025–2026

- Built a local LLM backend integrated with Unreal Engine 5 using WebSocket messaging, retrieval-augmented generation, per-NPC memory and JSON-schema validation; Unreal remained authoritative over game state.
- Delivered a packaged RPG prototype with dialogue, quests, inventory, save/load and text-to-speech; dialogue quality improved from 3.89 to 8.92/10 in qualitative evaluation.

Invoice Capture – B2B E-Invoicing Platform

Advanced Robotics and Computing

- Architecting invoice capture, validation and processing workflows for UK construction with a Python/TypeScript, React and PostgreSQL stack and AI-assisted document handling.

FPGA Racing – Multiplayer Unreal Engine 5 Game

C++ / FPGA / AWS GameLift

- Developed a multiplayer racing game in C++ and Unreal Engine 5, integrating an FPGA accelerometer controller with real-time game input and networking.

C Compiler and RISC-V CPU

Systems programming / computer architecture

- Implemented a C compiler source-to-output pipeline and designed a RISC-V CPU from instruction-set requirements to a working processor implementation.

Technical Skills

Languages: C++, Python, TypeScript, JavaScript, SQL Backend/Web: React, Node.js, Spring Boot, PostgreSQL, SQLite, REST APIs

Systems: Data structures and algorithms, computer architecture, compiler construction, real-time systems, WebSocket networking

AI: Local LLM integration, RAG, llama.cpp, applied machine learning

Tools/Cloud: Git, Firebase, Supabase, AWS GameLift, MQTT, CI/CD